

A convergent $o(1/k)$ series whose Poisson transform is not L^1 A full negative answer to Remark 3.4(c) of arXiv:2404.17289

Automated mathematics run `fa_banach.001`, lane 14

13 August 2026

Status. Candidate full counterexample, likely valid; human review recommended.

1 Source question

The source is A. K. J. Pritchard and D. Seifert, *The asymptotic behaviour of the Cesàro operator*, arXiv:2404.17289[1]. Given a sequence $(a_k)_{k \geq 0}$, it considers

$$f(t) = e^{-t} \sum_{k=0}^{\infty} a_k \frac{t^k}{k!}.$$

Remark 3.4(c), on printed page 6, asks whether convergence of $\sum_{k \geq 0} a_k$ implies $f \in L^1(0, \infty)$, even under the stronger assumption $a_k = o(k^{-1})$. It consequently leaves open whether the paper's series condition (3.2) is strictly weaker than its integral condition (3.1).

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- Remarks 3.4.** (a) For the purposes of deducing convergence of the series in (3.2) from (3.1) it would suffice to have Lemma 3.3 only for $a_k = o(k^{-1})$ as $k \rightarrow \infty$. In this case the application of Littlewood's theorem could be replaced by an application of Tauber's theorem [23].
- (b) The conclusion of Lemma 3.3 remains true under the much weaker conditions that the integral in (3.3) exists as an improper Riemann integral and $a_k = O(k^{-1/2})$ as $k \rightarrow \infty$; see [10, Thm. 126 & Thm. 156]. The proof of this result is considerably more involved than that of Lemma 3.3.
- (c) It follows, conversely, from [10, Thm. 122] that the integral in (3.3) exists as an improper Riemann integral whenever the series $\sum_{k=0}^{\infty} a_k$ is convergent, and in this case, too, (3.3) holds. There are large classes of sequences $(a_k)_{k \geq 0}$ for which it is even possible to show that $f \in L^1(0, \infty)$, and in particular this holds whenever $(a_k)_{k \geq 0} \in \ell^1$. However, we do not know whether this converse of Lemma 3.3 holds in general, even under the stronger assumption that $a_k = o(k^{-1})$ as $k \rightarrow \infty$. In particular, we leave open whether requiring convergence of the series in (3.2) is *strictly* weaker than (3.1).

We continue with an elementary yet important observation.

Lemma 3.5. *Let $T: c \rightarrow c$ be the Cesàro operator, and let $x = (x_k)_{k \geq 0} \in c$.*

Figure 1: Real full-width crop of Remark 3.4 from printed page 6.

2 Idea of the proof

A convergent series can be manufactured as the discrete derivative of any null sequence. We choose a null sequence that oscillates on logarithmic scales, so its discrete derivative has infinite total variation but changes negligibly across the $\sqrt{t}$ -wide window sampled by a Poisson random variable of mean t . Poisson smoothing therefore preserves the leading term $-\cos(\log t)/(t \log t)$. Its sign oscillates, so its improper integral converges as Hardy's theorem predicts, but its absolute integral diverges.

3 Counterexample

Fix $c = e^2$ and define, for $x \geq 0$,

$$s(x) = \frac{\sin(\log(x+c))}{\log(x+c)}, \quad h(x) = s(x) - s(x+1), \quad a_k = h(k).$$

Lemma 1 (Poisson smoothing). *If N_t has the Poisson distribution with mean t , then*

$$\mathbb{E}h(N_t) = h(t) + O\left(\frac{1}{t^2 \log t}\right) \quad (t \rightarrow \infty).$$

Proof. Direct differentiation, with $y = x + c$, gives

$$s^{(j)}(x) = O\left(\frac{1}{y^j \log y}\right), \quad j = 1, 2, 3.$$

Since $h''(x) = s''(x) - s''(x+1)$, the mean-value theorem yields

$$h''(x) = O\left(\frac{1}{(x+c)^3 \log(x+c)}\right). \quad (1)$$

Let $E_t = \{|N_t - t| \leq t/2\}$. On E_t , Taylor's theorem at t and (1) give

$$h(N_t) = h(t) + h'(t)(N_t - t) + O\left(\frac{(N_t - t)^2}{t^3 \log t}\right).$$

Now $\mathbb{E}(N_t - t) = 0$ and $\mathbb{E}(N_t - t)^2 = t$. Standard Poisson Chernoff bounds give $\mathbb{P}(E_t^c) \leq 2e^{-\gamma t}$ for an absolute $\gamma > 0$. Since h is bounded and $h'(t) = O(t^{-2}/\log t)$, Cauchy–Schwarz bounds all terms created by restricting the constant and linear terms to E_t by an exponentially small quantity. Taking expectations therefore gives

$$\mathbb{E}h(N_t) - h(t) = O\left(\frac{\mathbb{E}(N_t - t)^2}{t^3 \log t}\right) + O(e^{-\gamma t/2}) = O\left(\frac{1}{t^2 \log t}\right).$$

□

Theorem 1. *The series $\sum_{k=0}^{\infty} a_k$ converges and $a_k = o(k^{-1})$, but*

$$F(t) := e^{-t} \sum_{k=0}^{\infty} a_k \frac{t^k}{k!} \notin L^1(0, \infty).$$

More precisely,

$$F(t) = -\frac{\cos(\log(t+c))}{(t+c) \log(t+c)} + O\left(\frac{1}{t(\log t)^2} + \frac{1}{t^2 \log t}\right). \quad (2)$$

Proof. The sum telescopes:

$$\sum_{k=0}^n a_k = s(0) - s(n+1) \longrightarrow s(0).$$

Moreover,

$$s'(x) = \frac{\cos(\log(x+c))}{(x+c)\log(x+c)} - \frac{\sin(\log(x+c))}{(x+c)(\log(x+c))^2}.$$

Since $h(x) = -\int_x^{x+1} s'(u) du$ and $s''(x) = O((x+c)^{-2}/\log(x+c))$, we have

$$h(x) = -\frac{\cos(\log(x+c))}{(x+c)\log(x+c)} + O\left(\frac{1}{x(\log x)^2} + \frac{1}{x^2 \log x}\right). \quad (3)$$

In particular, $a_k = O(k^{-1}(\log k)^{-1}) = o(k^{-1})$.

The defining series for F is the Poisson expectation $F(t) = \mathbb{E}h(N_t)$. Combining Lemma 1 with (3) proves (2).

Put $u = \log(t+c)$. For all sufficiently large t satisfying $|\cos u| \geq 1/2$, the error in (2) is at most $1/(4(t+c)u)$, and hence

$$|F(t)| \geq \frac{1}{4(t+c)u}.$$

The set $\{u : |\cos u| \geq 1/2\}$ contains intervals of a fixed positive length in every period. Since $dt/(t+c) = du$,

$$\int_{\substack{t \text{ large} \\ |\cos \log(t+c)| \geq 1/2}} \frac{dt}{(t+c)\log(t+c)} = \int_{\substack{u \text{ large} \\ |\cos u| \geq 1/2}} \frac{du}{u} = \infty.$$

Thus $F \notin L^1(0, \infty)$. □

4 Exact consequence for the Cesàro conditions

Define a convergent sequence $x = (x_k)_{k \geq 0}$ by

$$x_0 = 0, \quad x_{k+1} = (k+1)a_k \quad (k \geq 0).$$

The estimate on a_k shows $x_k \rightarrow 0 = x_0$. Furthermore,

$$\sum_{k=1}^{\infty} \frac{x_k - x_0}{k} = \sum_{j=0}^{\infty} a_j$$

converges, so condition (3.2) of the source holds. But its condition (3.1) has integrand

$$e^{-t} \left| \sum_{k=1}^{\infty} (x_k - x_0) \frac{t^{k-1}}{k!} \right| = e^{-t} \left| \sum_{j=0}^{\infty} a_j \frac{t^j}{j!} \right| = |F(t)|,$$

whose integral diverges. Hence (3.2) is strictly weaker than (3.1), fully answering the second sentence of Remark 3.4(c).

5 Verification, limitations, and novelty

The accompanying numerical script verifies the telescoping identity, the Poisson-smoothing error, and the asymptotic sign on logarithmic periods. It is not used in the proof. A bounded search of the run indexes, exact source phrases, the title/citations, and close Poisson-transform variants through 2026-08-13 found no later answer or matching construction. The search was not an exhaustive bibliographic review, so novelty remains subject to human check.

References

- [1] A. K. J. Pritchard and D. Seifert, *The asymptotic behaviour of the Cesàro operator*, J. Math. Anal. Appl. **552** (2025), 129748; arXiv:2404.17289.